# CST511 - Computer Networks

## CST 511 (Computer Networks) Syllabus

### Course Information

#### Course Details

- **Course Title:** Computer Networks
- **Course Number:** CST 511
- **Meeting Location:** Physical
- **Units:** 2

#### Course Description

This course covers fundamental networking concepts, including Local Area Networks (LANs), Wide Area Networks (WANs), and key internet protocols such as TCP/IP. Emphasis is placed on practical applications, network design, and the latest trends in modern networking technologies. Through hands-on activities and case studies, students will explore real-world networking scenarios and emerging developments in the field.

#### Materials

- Course textbook: "Computer Networks: A Top-Down Approach" by James Kurose and Keith Ross

#### Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

#### Course Objectives

At the end of this class, you should be able to:

1. Understand and explain fundamental networking concepts and protocols including TCP/IP
2. Design and analyze Local Area Networks (LANs) and Wide Area Networks (WANs)
3. Implement and troubleshoot network configurations in practical scenarios
4. Evaluate emerging networking technologies and their real-world applications

### Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

| Assignment Kind | Value |
|---|---|
| Projects | 35% |
| Exams | 35% |
| Participation | 30% |

### Projects

Projects will take the form of network configuration exercises, protocol analysis assignments, and basic network design challenges. Students will gain hands-on experience with fundamental networking concepts through practical labs and focused implementation work.

### Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

### Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

### Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

| Grade Range | Letter Grade |
|---|---|
| $[97, \infty)$ | A+ |
| $[93, 97)$ | A |
| $[90, 93)$ | A- |
| $[87, 90)$ | B+ |
| $[83, 87)$ | B |
| $[80, 83)$ | B- |
| $[77, 80)$ | C+ |
| $[73, 77)$ | C |
| $[70, 73)$ | C- |
| $[60, 70)$ | D |
| $[0, 60)$ | F |

## Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.